MCBs (Miniature Circuit Breaker)



MCBs have been replacing electric fuse from wirings at most of the places. The electric fuse has a big practical problem. Whenever the wire fuses, one needs to replace the wire to resume electric supply. More often than not, this proves to be a cumbersome task. Miniature circuit breakers break the circuit automatically. One just needs to switch it on to resume the electric supply. Many models of MCBs have a built in mechanism by which the electric supply is automatically resumed.

2.8. Magnetic Effect of electricity

The next effect of electric current is Magnetism. In 1819, Hans Christian Oersted discovered the electricity that has a magnetic effect. The experiment in activity-5 will help you understand the magnetic effect of electric current.

2.8.1. Application of magnetic effect of electric current - Electromagnet

Magnetic effect of electric current has been used in making powerful electromagnets. Electromagnets are also used to remove splinters of steel or iron in hospitals dealing with eye injuries.

Electro magnets are used in many appliances that we use in our day to day life, namely, electric bell, cranes and telephone. Let us know how the magnetic effect of electric current is applied in telephones.

ACTIVITY 5

Materials required

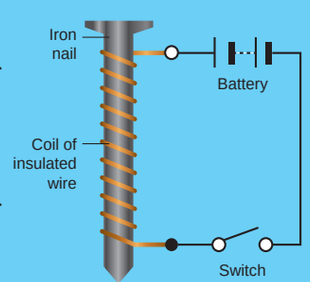
- Iron nail
- Battery & Switch
- Wire

Take around 75 cm long piece of insulated flexible wire and an iron nail say about 8 - 10 cm long. Wind the wire tightly around the nail in the form of a coil. Connect the free ends of the wire to the terminals of a cell as shown. Place some pins on or near the end of the nail. Now switch on and switch off the current, What happens?

When the switch is kept in on position the pins starts to cling to the end of the nail.

When the electric current is switched off the coil generally loses its magnetism. Such coils are called as electromagnets.

The polarities of both ends of the coil changes according to the direction of electric current passes.



2.8.2. Telephone

In telephones, a changing magnetic effect causes a thin sheet of metal (diaphragm) to vibrate. The diaphragm is made up a metal that can be attracted to magnets.

1. The diaphragm is attached to spring that is fixed to the earpiece.
2. When a current flows through the wires, the soft – iron bar becomes an electromagnet.

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